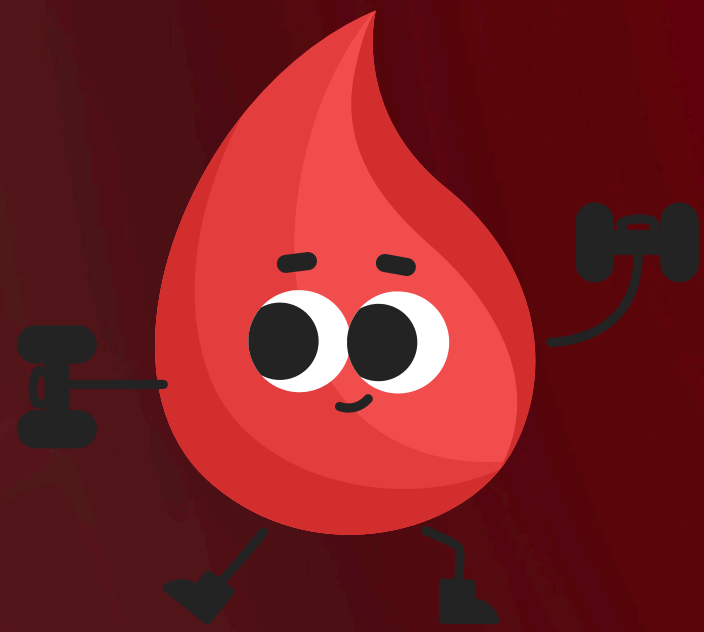


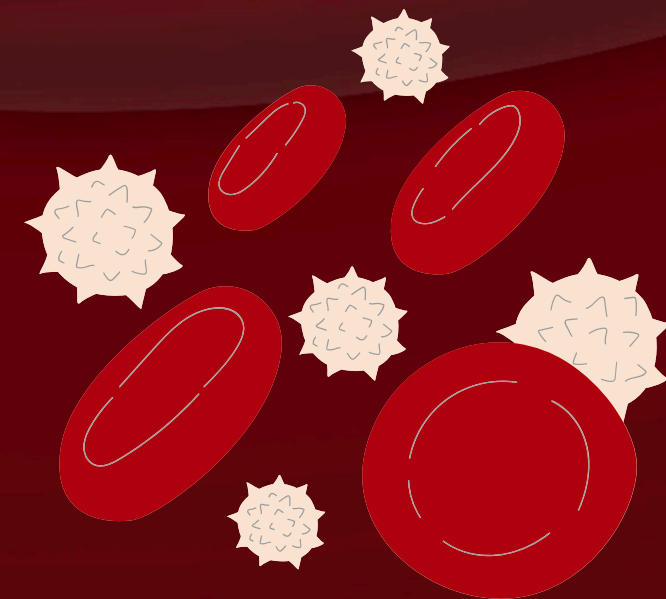


SMART BLOOD CELL CLASSIFICATION: THE INTERSECTION OF AI AND MEDICINE

BY: LUPE COVARRUBIAS

PROBLEM STATEMENT

By automating blood cell classification, this initiative leverages AI to accelerate diagnosis, enhance healthcare efficiency, and pave the way for faster, more accurate disease detection and treatment.



HOW CAN WE HELP?

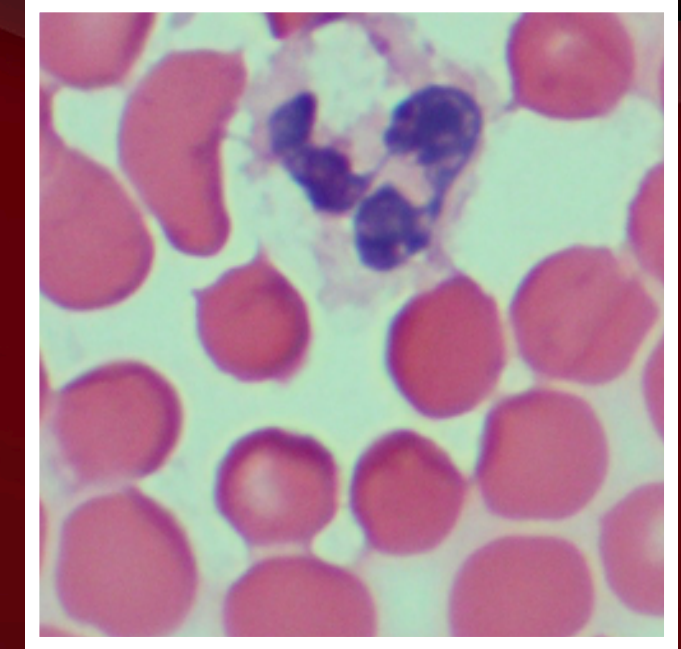
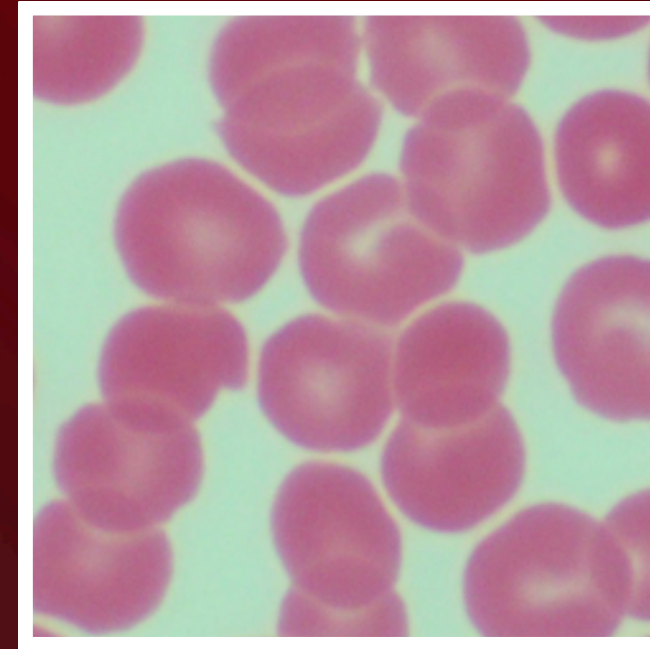


The project aims to address the limitations of manual blood cell classification, which is time-consuming and prone to human error, by leveraging the power of deep learning to create a more efficient and accurate solution.

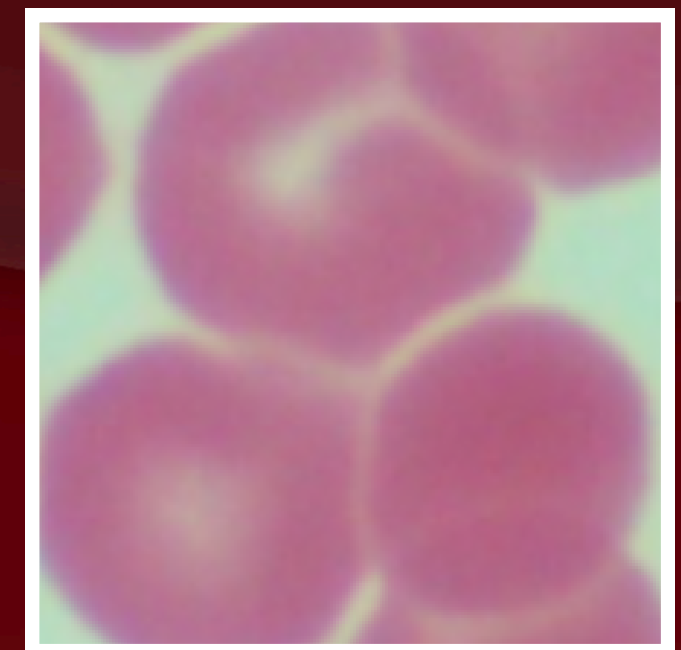
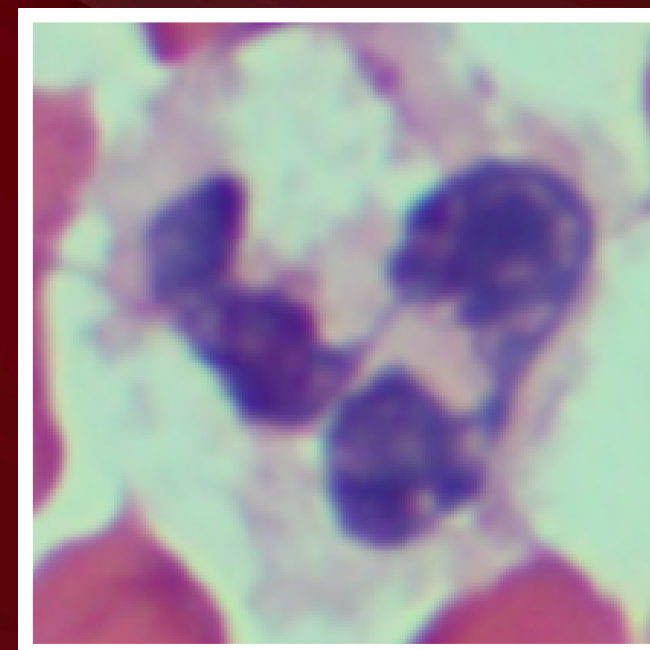


DATA OVERVIEW

- Dataset sourced from Kaggle.com and edited to conform to binary classification
- (100, 2)
- 49 white blood cell (wbc) images
- 51 red blood cell images (rbc)
- (224, 224)



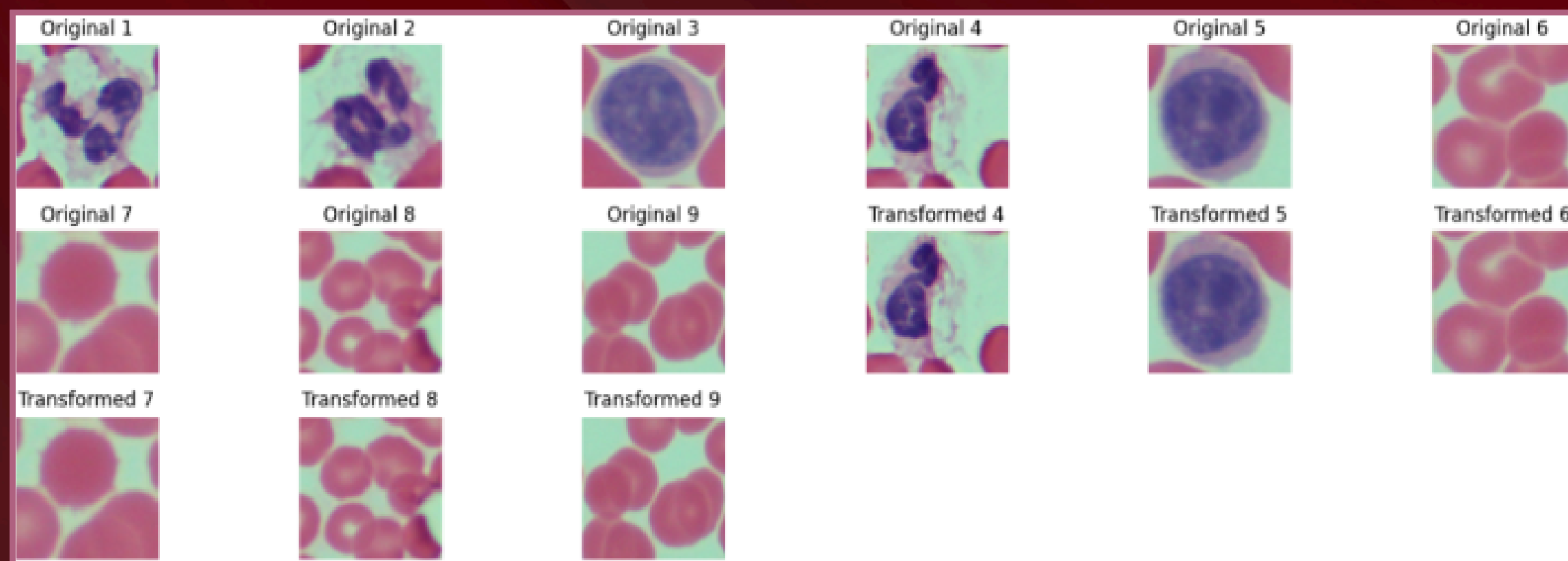
Figures 1 - 2: Examples of wbc images before cropping



Figures 3 - 4: Examples of rbc images after cropping

PRELIMINARY DATA AUGMENTATION

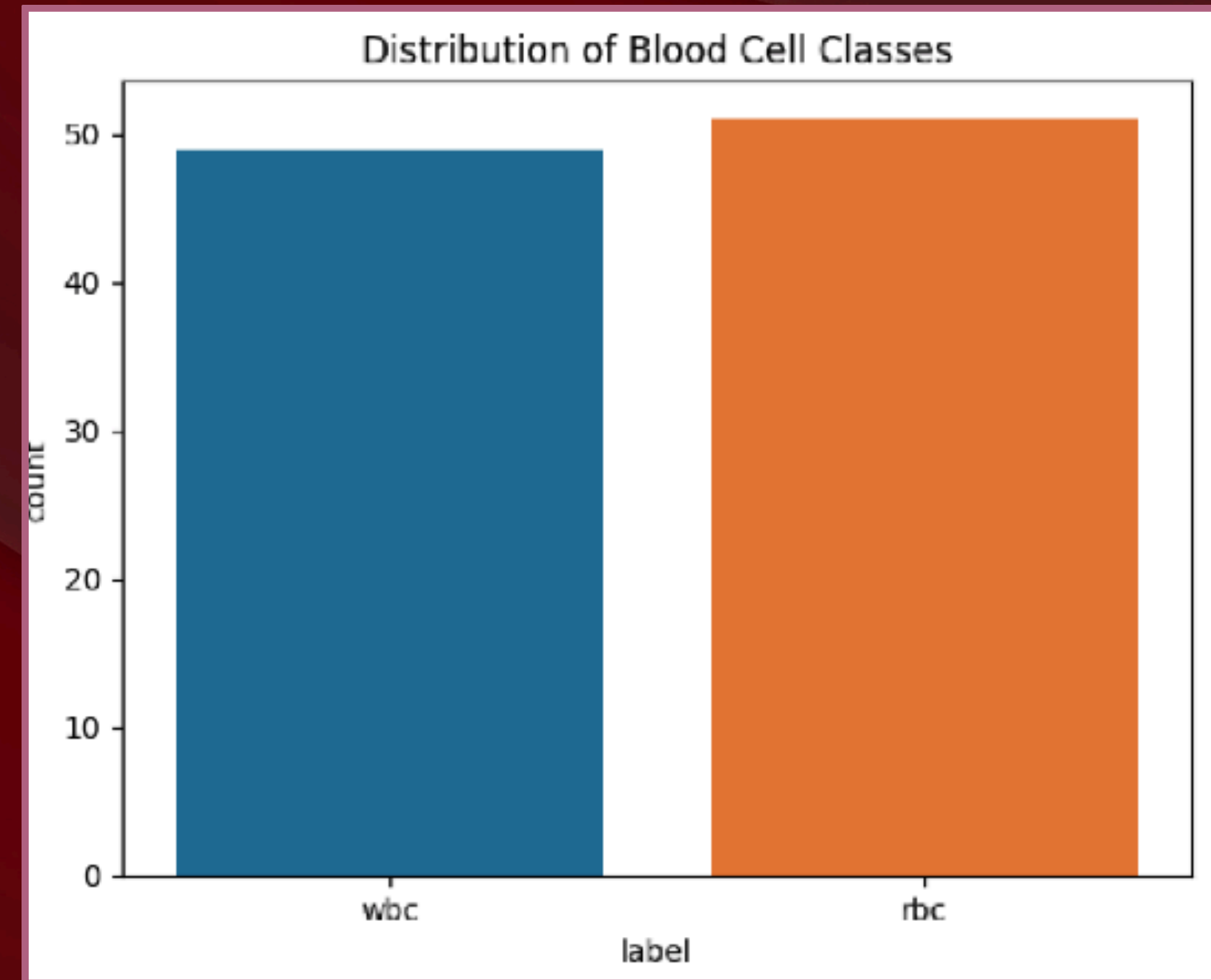
Unsharp masking technique: enhance edges
for better feature extraction



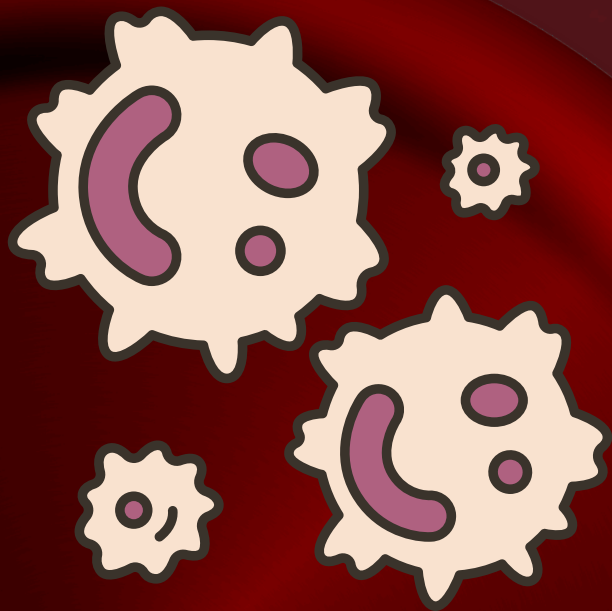
Figures 5: Snippet of images before and after applying unsharp masking technique

EXPLORATORY DATA ANALYSIS

- (224, 224, 3)
- balanced dataset
- updated file path
- updated csv



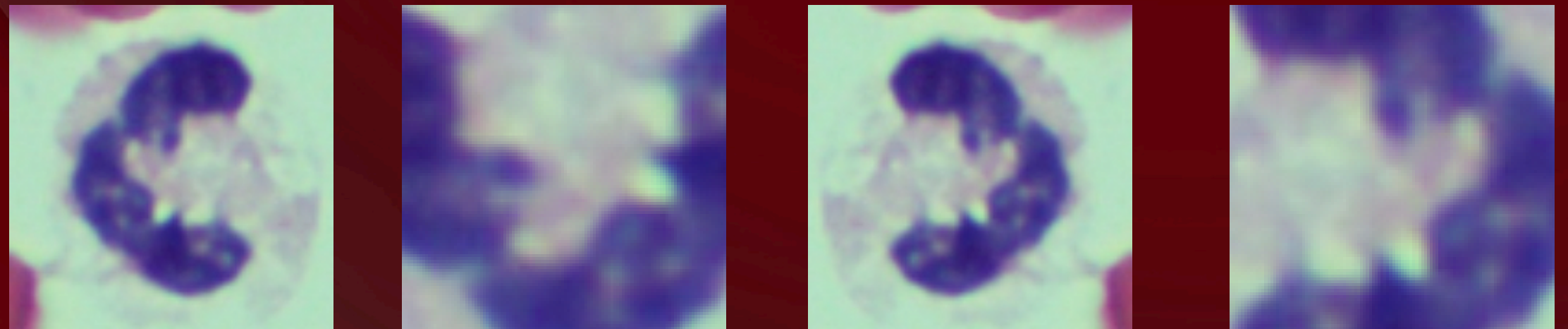
Figures 6: Seaborn countplot of the distribution of blood cell classes



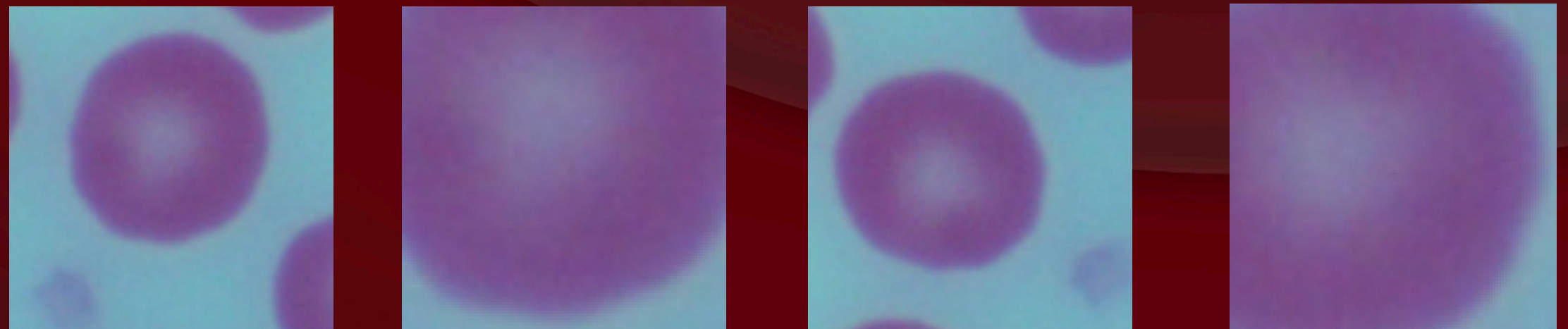
PREPROCESSING

DATA AUGMENTATION

- increase data size
- flipping
- cropping
- rotating
- brightness
- (1600, 2)
- rbc: 0
- wbc: 1



Figures 7: Seaborn countplot of the distribution of blood cell classes



Figures 8: Seaborn countplot of the distribution of blood cell classes

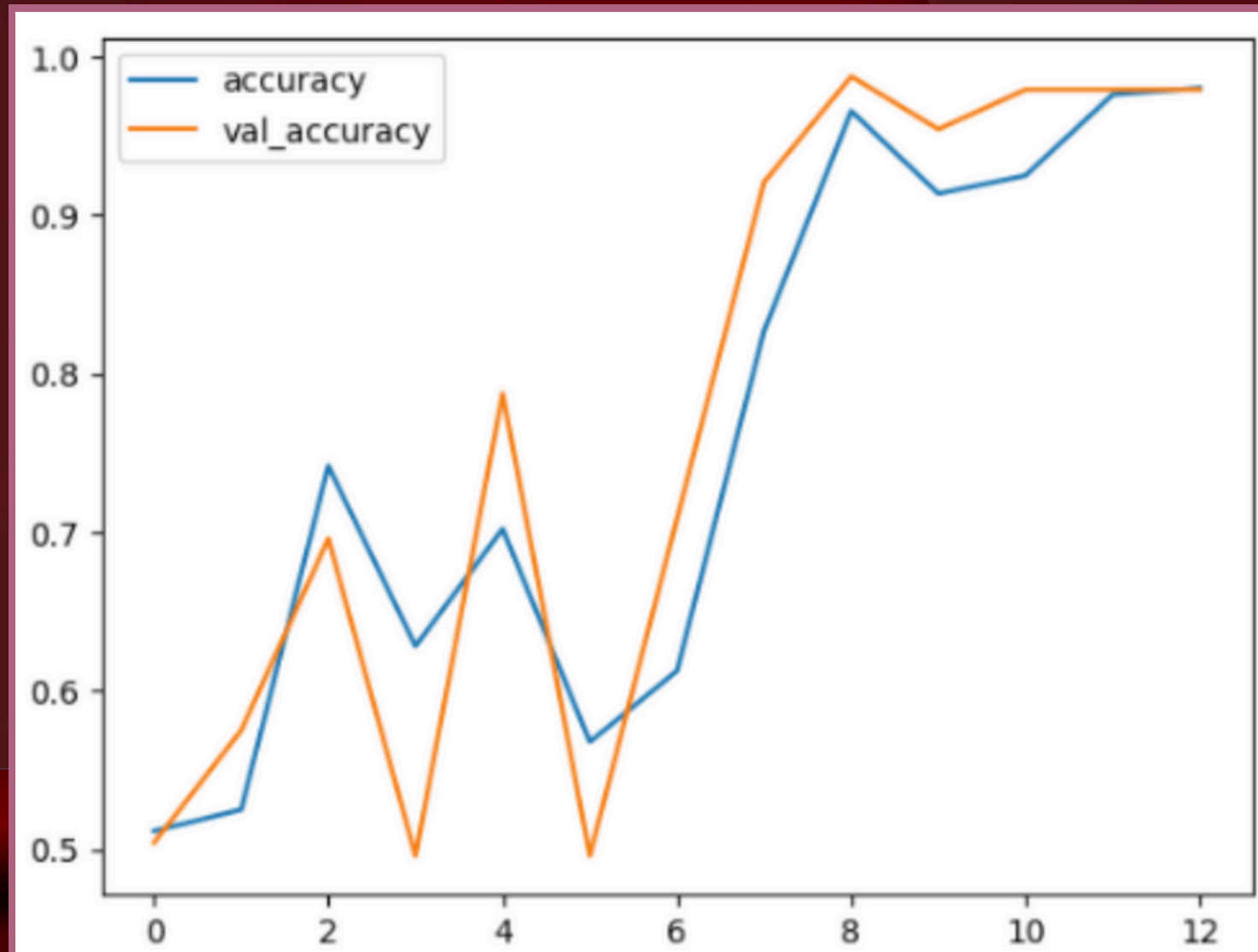
CUSTOM CNN MODEL

- consistent problem: vanishing gradient
- activation function: ELU
- MaxPool2D
- gradual increase in neurons
- Sigmoid function
- test size: 25%
- optimizer: RMSprop
- binary cross-entropy
- moderate to high learning rate
 - scheduled learning rate
- gradient clipping
- momentum
- 20 epochs
- batch size: 32
- validation size: 20%

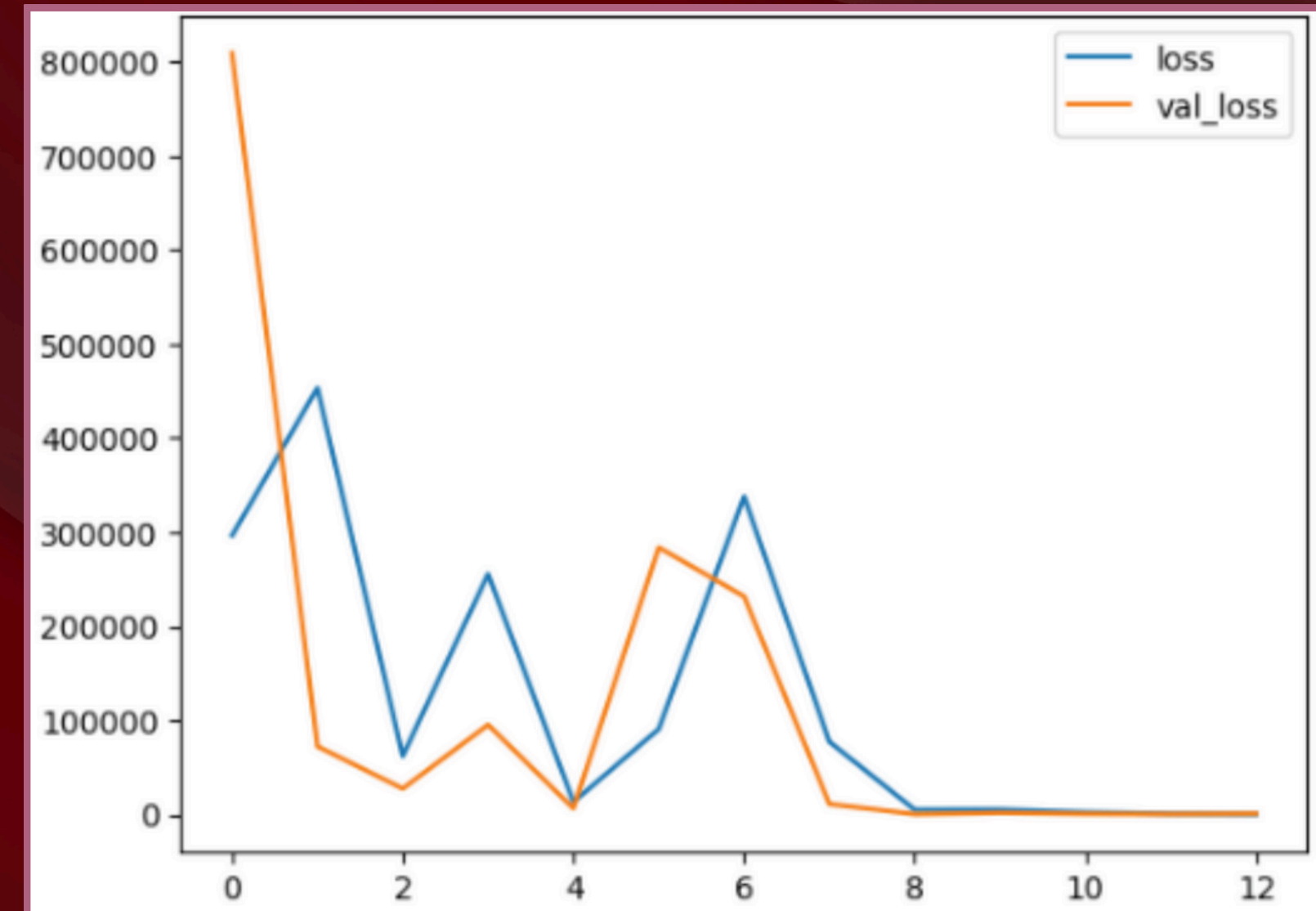
	precision	recall	f-1 score
0	0.97	1.00	0.98
1	1.00	0.97	0.98
accuracy			0.98



CUSTOM CNN MODEL



Figures 9: Custom model accuracy values at each epoch for training and validation data



Figures 10: Custom model loss function values at each epoch for training and validation data



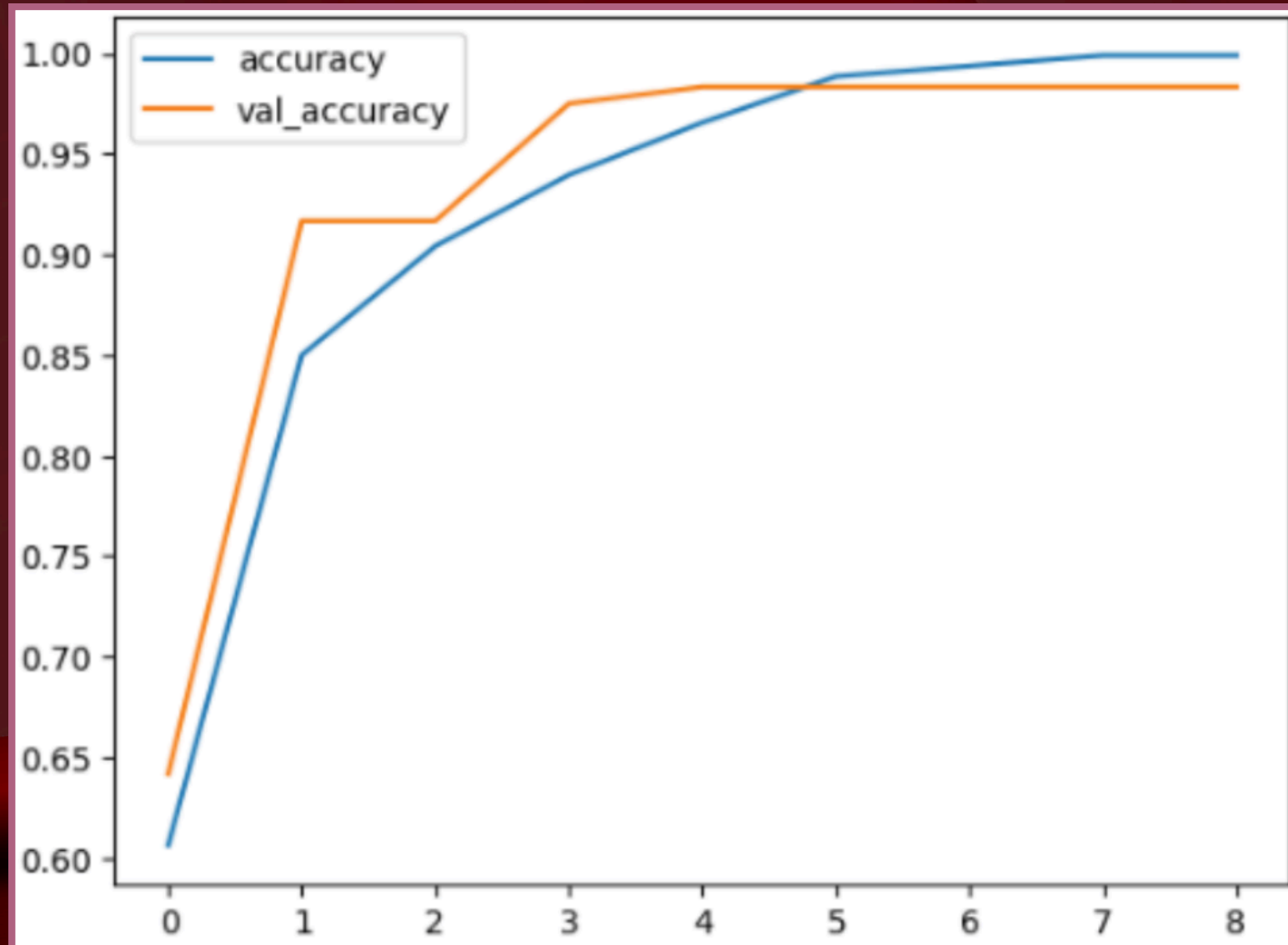
PRE-TRAINED CNN MODEL



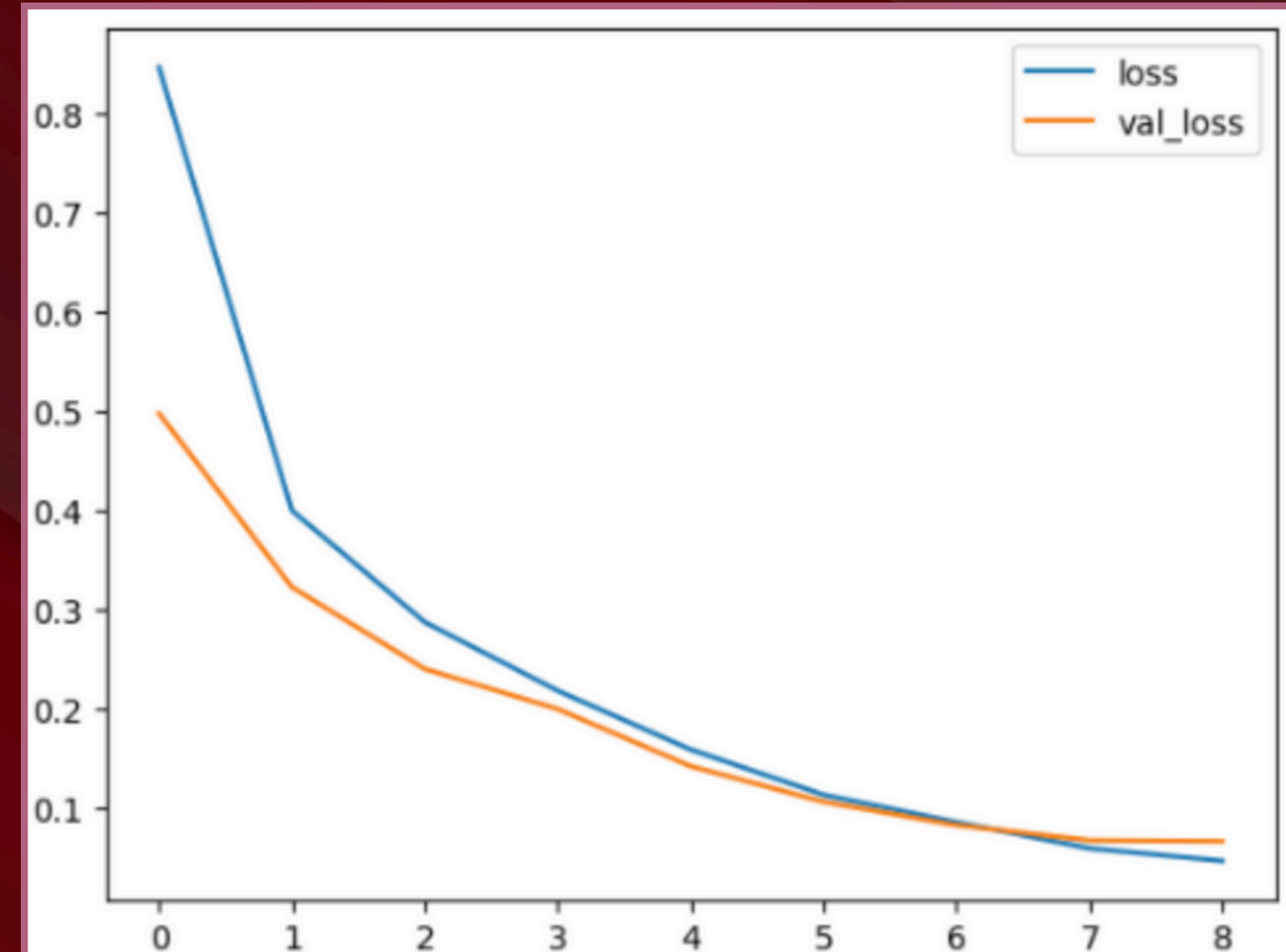
- VGG16
- convolutional layers frozen
- 128 neurons
- ReLu activation
- sigmoid activation
- optimizer: Adam
- binary cross-entropy
- 10 epochs
- batch size: 32
- validation split: 20%

	precision	recall	f-1 score
0	0.98	1.00	0.99
1	1.00	0.98	0.99
accuracy			0.99

PRE-TRAINED CNN MODEL



Figures 11: Pre-trained model accuracy values at each epoch for training and validation data



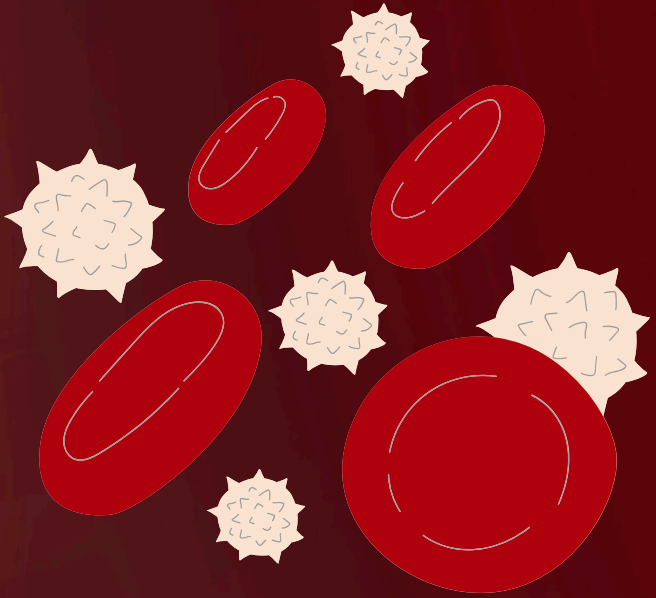
Figures 12: Pre-trained model loss function values at each epoch for training and validation data

MODEL COMPARISON

specific requirements,
such as medical
imaging, for task
specificity and
control, might be a
viable alternative

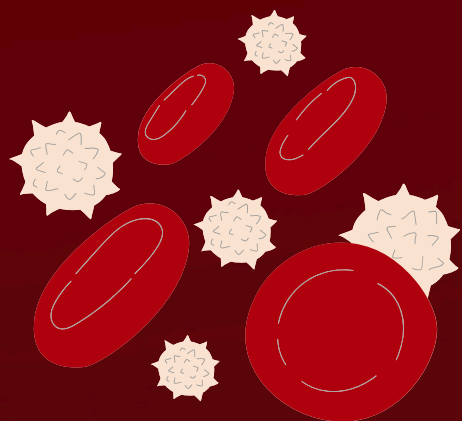
Exceptional
performance
strong generalization
high accuracy

significantly
accelerates
learning, improves
generalization,
and leads to better
overall
performance



CONCLUSION

- potential to improve patient care, reduce costs, and drive innovation
- leverage automation leading to lower labor costs for healthcare providers
- high accuracy of the model can minimize errors in blood cell analysis
- faster results can then reduce patient anxiety
- can be expanded to new markets and applications, such as veterinary medicine or environmental monitoring
- potential improvements: classification of different types of white blood cells



THANK
YOU

